

Norbert Muzila

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ABOUT ME

Space systems engineer with a data science background, building intelligent, autonomous space systems and the software that flies and operates them. Works across the whole chain: MBSE and mission design, onboard and ground segment software, Earth observation and GeoAI pipelines, and the operations dashboards on top. Hands-on with CubeSats, rovers, UAVs and robotic systems, from subsystem integration and testing to autonomous navigation. Used to multidisciplinary international teams and tight deadlines.

WORK EXPERIENCE

Chief Technology & Strategy Officer

Jul 2026 to Present

Space4Climate · Voluntary · Remote

- Set the technical strategy and engineered a secure cloud document database with granular access governance.
- Programmed a centralised intelligence platform, integrated with Discord, that automates team workflows and communications.
- Direct the engineering pipelines that turn complex satellite data into AI-driven public climate media.

AI & Earth Observation Research and Innovation Intern

Apr 2026 to Present

LUNEX EuroMoonMars · Netherlands

- Earth observation and 3U CubeSat AI/SW subsystem lead for autonomous onboard data processing.
- Developing a multi-task AI framework for integrated urban monitoring across global smart cities.
- Fusing Copernicus EO imagery with near real-time IoT sensor telemetry into a single operational picture.

Sponsorship Manager, Organising Team

2026

Space Generation Advisory Council · SG[France] 2026

- Secured over €8,000 in sponsorship for SGAC's flagship French event, negotiating and executing agreements with corporate partners under the Executive Director and BDO Officer.

Contributor, Open Science Working Groups

2025 to Present

NASA Open Science Data Repository (OSDR) · Voluntary

- Technical contributor to two focus groups: Artificial Intelligence & Machine Learning, and Hardware.

Data Science Resident Researcher

2024 to 2025

Zimbabwe Centre for High Performance Computing · Harare, Zimbabwe

- Engineered scalable Python ETL pipelines and ran statistical time-series analysis on large-scale operational datasets.
- Optimised machine learning algorithms for HPC and cloud environments, supporting data-driven decisions at C-level.
- Awarded the Certificate of Competence in AI and Big Data Analytics.

Strategy, Digital Transformation & DevOps Intern

2023 to 2024

Steward Bank · Harare, Zimbabwe

- Built machine learning and deep neural network models for customer churn prediction, improving account retention by 88%.
- Automated containerised CI/CD pipelines with Docker and GitLab, cutting deployment time by 90%.
- Raised development efficiency by 70% through Elasticsearch log analysis and real-time Kibana dashboards; built Power BI reporting for executives.

PROJECTS

ARGUS — Maritime GNSS Resilience & Forensic Intelligence

ESA SBIC · Ongoing

Founder & Developer · argusgrid.vercel.app

- Early-warning and forensic platform for maritime GNSS interference across European and international waters.
- Delivers 12 to 24 hour advance interference warnings and legally admissible forensic dossiers, protecting commercial fleets against GPS spoofing, jamming and navigation disruption.

Rover and Drone Duo for Subsurface Seismic Exploration

2025 to 2026

AI & Software Lead · International Space University team project

- Led the AI and software subsystem of a 20-person Mars lava tube mission concept, designing the autonomous navigation and exploration algorithms. Accepted for presentation at IAC 2026.

1U CubeSat, Project-Based Learning

2025 to 2026

Project Lead & Software Subsystem Lead · International Space University

- Coordinated a team of 8 across integration scheduling, interface management and design reviews.

- Led full hardware integration to CubeSat Design Specification standards: EPS, OBDH, UHF/VHF communications, ADCS sensors and structural assembly.

Urban Monitoring Platform & ISU Groundstation Command Deck

2026

norbertmuzila.github.io

- Multi-task AI framework fusing Copernicus EO imagery with IoT sensor networks for smart city monitoring.
- Real-time mission operations cockpit: serial telemetry unpacking, SGP4 orbit propagation, 2D Leaflet maps with Esri imagery, 3D Globe.gl orbit prediction and wireless camera downlink.

ABIL — Environmental Scoring Platform

2025 to Present

Co-developer · clearesg.vercel.app

- Condenses satellite readings into a single environmental score per site for risk, insurance, investment and ESG, with every score traced back to its measurement and the regulation it answers to.

Also ISU Robotics Competition (Chief Programmer, autonomous navigation and gemstone collection in Python), the ESA Member States Space Agency Atlas (interactive GIS catalogue of all 22 member nations), the Umbrella Seismic Sensing Model, and published research on adversarial machine learning and AI security in autonomous systems.

EDUCATION AND TRAINING

Master of Space Studies (MSS26) — Space Engineering & Applications

2025 to 2026

International Space University · Illkirch-Graffenstaden, France

- Space systems engineering, astrodynamics, satellite communications, orbital mechanics, MBSE, mission planning and remote sensing.
- Hands-on development, integration and testing of CubeSats, rovers, UAVs and autonomous robotic systems across avionics, flight control, telemetry and simulation.

BSc (Honours) in Data Science & Systems

2020 to 2024

University of Zimbabwe · Harare, Zimbabwe

- Machine learning, deep neural networks, big data analytics, predictive modelling, and software systems design and development.

Certificates: Executive Space Course, ISU (2025) · AI & Big Data Analytics, ZCHPC (2025) · AWS Cloud Practitioner · Google Advanced Data Analytics · Cisco Cybersecurity Essentials.

SKILLS

Space systems MBSE and Cameo SysML, orbital mechanics and astrodynamics, STK, mission operations, flight software, AIT, verification and validation, CubeSat subsystems (EPS, OBDH, ADCS, UHF/VHF).

Earth observation & geospatial Copernicus and Sentinel data, remote sensing, Google Earth Engine, QGIS, digital twins, geospatial visualisation with Leaflet.js.

AI & data TensorFlow, PyTorch, scikit-learn, deep learning, NLP, ETL pipelines, time-series and statistical analysis, Power BI, Grafana.

DIGITAL SKILLS

Programming Python, C++, MATLAB, JavaScript, SQL, R

Tools & platforms Docker, Git and GitLab, CI/CD, Linux, Elasticsearch and Kibana, Jupyter, Pandas, AWS, SolidWorks, Fusion 360

LANGUAGE SKILLS

Mother tongue Shona

English C2, full professional proficiency, used daily in study, research and technical writing

French A2, basic user, currently living and studying in France

INTERESTS

Chess, darts and table tennis: three games that reward patience, precision and a steady hand when the pressure is on, and the reason most of my downtime happens across a board or a table.